Review: Exponents and radicals

1 As the base is 4 and the power is 5, you multiply 4 by itself 5 times. So $4^5 = 4 \times 4 \times 4 \times 4 \times 4$

2 Check the power. If the power is odd then the answer will be negative because a negative multiplied by itself an odd number of times will give you a negative. So in this case, for $(-7)^3$, the answer is negative.

3

a No

b No

c Yes

d Yes

4 **B**

5

a 3

b 1

c 64

d 16

e 243

f -64

g -648

h -216

i 45

j 40

k 36

l -64

m 81

n -81

o 16

p 256

q -225

r -64

s -324

t 17

u -5

v 3

w 37

x -63

y 91

z -158

6 $39^2\ \mathrm{m}^2$

7

a True

b False

c False

8

- **a** 2
- **b** 4
- **c** 8
- **d** 10
- **e** 12
- **f** 2
- **g** 10
- **h** 8
- **i** 8
- **j** 4
- **k** 10
- **l** 13
- **m** 15
- **n** -2
- **o** -9
- **p** -11
- **q** -14

Additional questions

9

- **a** 5
- **b** 7
- **c** -6
- **d** -7

10

- **a** 2
- **b** 4
- **c** -3
- **d** -5

11

- **a** 2
- **b** 4
- **c** 5
- **d** 3
- **e** -1
- **f** -3
- **g** -2
- **h** -4

12 3^3

13

- **a** 31
- **b** 49
- **c** 80
- **d** 64
- **e** 125
- **f** 3
- **g** 4
- **h** 72
- **i** $\frac{3}{4}$
- **j** $\frac{8}{9}$
- **k** 1
- **l** 1
- **m** $\frac{4}{5}$
- **n** 1
- **o** $\frac{9}{5}$
- **p** 9

14

a

i Ralph distributed the parentheses before taking the square.

ii $2\,(3+2)^2 = 2\,(5)^2 = 2 * 25 = 50$

b

i Ida moved the (9) into the square root without squaring it first.

ii $\sqrt{9}\left(8 \div 4 - 3 \times 4 + 4^2 + 3\right) = \sqrt{9}\,(2 - 12 + 16 + 3) = \sqrt{9}\,(9) = 3\,(9) = 27$